INDUSTRIAL ORIENTED MINI PROJECT

On

MULTIPLATFORM REPUTATION ANALYSIS SYSTEM

Submitted in fulfillment of the requirements for the award of the degree of

Bachelor of Technology

In

Information Technology

By

SHAHISTA TAMKEEN - 20261A1249

Under the guidance of

Mrs.U.CHAITANYA

Assistant Professor, IT

DEPARTMENT OF INFORMATION TECHNOLOGY

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY

(AUTONOMOUS)

(Affiliated to JNTUH,Hyderabad; Eight UG Programs Accredited by NBA;

Accredited by NAAC with 'A++' Grade)

Gandipet, Hyderabad, Telangana, Chaitanya Bharati (P.O),

Ranga Reddy District, India – 500075.

2023-2024

CERTIFICATE

This is to certify that the Project entitled **Multiplatform Reputation Analysis System** submitted by **Shahista Tamkeen - 20261A1249**in partial fulfillment of the requirements for the Award of the Degree of Bachelor of Technology in Information Technology as specialization is a record of the bonafide work carried OUT under the supervision **of Mrs. J Hima Bindu**, and this has not been submitted to any other University or Institute for the award of any degree or diploma.

Project Guide:
Mrs.U.CHAITANYA
Assistant Professor
Dept. of IT.

Project Coordinator:
Mrs. J HIMABINDU
Assistant Professor
Dept. of IT

External Examiner:

DR. D. VIJAYA LAKSHMI
Professor & HoD
Dept. of IT

DECLARATION

I hereby declare that the research work entitled **Multiplatform Reputation Analysis System** is original and bonafide work carried out by us as a part of fulfilment for Bachelor of Technology in Information Technology, Mahatma Gandhi Institute of Technology, Hyderabad, under the guidance of Mrs.U.Chaitanya , Assistant Professor, Dept. of IT, MGIT.

No part of the work is copied from books/journals/internet and wherever the portion is taken, the same has been duly referred in the text. The report is based on the work done entirely by me and not copied from any other source.

SHAHISTA TAMKEEN - 20261A1249

ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of any task would be incomplete without introducing the people who made it possible and whose constant guidance and encouragement crowns all efforts with success. They have been a guiding light and source of inspiration towards the completion of the project.

I would like to express our sincere gratitude and indebtedness to our project guide **Mrs.U.Chaitanya**, Assistant Professor, Dept. of IT, who has supported us throughout our project with immense patience and expertise.

I are also thankful to the honorable Principal of **MGIT Prof. G. CHANDRA MOHAN REDDY** and **Dr. D. VIJAYA LAKSHMI**, HoD and professor, Department of IT, for providing excellent infrastructure and a conducive atmosphere for completing this project successfully.

I am also extremely thankful to our Project Coordinator **Mrs. J Hima Bindu**, Assistant Professor, Dept. of IT, for her valuable suggestions and interest throughout the course of this project.

I convey our heartfelt thanks to the lab staff for allowing us to use the required equipment whenever needed.

Finally, I would like to take this opportunity to thank our families for their support all through work. I sincerely acknowledge and thank all those who gave directly or indirectly their support in completion of this work.

SHAHISTA TAMKEEN - 20261A1249

ABSTRACT

This cutting-edge project leverages deep learning techniques for aspect-based sentiment analysis, revolutionizing the assessment of product and service reputation across diverse online platforms. By seamlessly gathering and processing data from various sources, it offers a holistic view of entity reputation. Notably, the system incorporates spam detection algorithms, enhancing the accuracy of sentiment analysis.

The innovation lies in its ability to calculate numerical reputation values for entities and their specific aspects, providing a quantitative measure of reputation. This approach transcends previous models by ensuring comprehensive evaluations, significantly improving precision and reliability in online reputation assessments. By adopting a nuanced perspective through aspect-based sentiment analysis, the project delivers more granular insights into the strengths and weaknesses of entities, contributing to a more informed decision-making process for consumers and businesses alike. This sophisticated methodology represents a significant step forward in the realm of reputation management, aligning with the ever-evolving dynamics of online platforms and the growing need for nuanced sentiment analysis.

LIST OF FIGURES

Figure 4.2.1 Use Case Diagram for Multiplatform Reputation Analysis System 17

Figure 4.2.2 Sequence Diagram for Multiplatform Reputation Analysis System 18

Figure 4.2.3 Class Diagram for Multiplatform Reputation Analysis System 19

Figure 4.4.4 Activity Diagram for Multiplatform Reputation Analysis System 20

Figure.5.3.1 Testing to upload Dataset 30

Figure 5.3.2 Testing to create a text length 30

Figure 5.3.3 Testing for review counts using stars 31

Figure 5.3.4 Testing for correlation between features 31

Figure 5.4.1 Output of Navie Bayes Algorithm 33

Figure 5.4.2 Output of Logistic Algorithm 33

Figure 5.4.3 Output of SVM Algorithm 34

Figure 5.4.4 Output of K-nearest neighbour Algorithm 34

Figure.5.4.5 Output for the Overall Comparison of Algorithms 35

LIST OF TABLES

Table.5.4.1 Output values for Navie Bayes Algorithm 38

Table.5.4.2 Output values for Logistic Regression Algorithm 39

Table.5.4.3 Output values for SVM Regression Algorithm 40

Table.5.4.4 Output values for K-nearest Algorithm 41

Table.5.4.6 Overall comparison of Algorithms 42

TABLE OF CONTENTS

Certificate

Declaration **i**

Acknowledgement **ii**

Abstract **iii**

List of Figures **iv**

1. Introduction **1**

1.1 Motivation 1

1.2 Problem Statement 2

1.3 Project Objectives 2

1.4 Project Report Organization 3

2. Literature Survey 4

2.1 Existing Work 6

2.2 Limitations of Existing Work 6

3. Software and Hardware Requirements 8

3.1 Functional Requirements 6

3.2 Non-Functional Requirements 6

4. System Analysis 10

4.1 Modules 10

4.2 Algorithms Used 11

4.2.1 Navie Bayes Algorithm 12

4.2.2 Logistic Algorithm 12

4.2.3 SVM Algorithm 13

4.2.4 K-nearest Neighbor Algorithm 14

4. System Design 15

4.1 Proposed methods 15

4.2 UML Diagrams 16

4.2.1 Use Case Diagram 16

4.2.2 Class Diagram 17

4.2.3 Sequence Diagram 18

4.2.4 Activity Diagram 20

5. Implementation And Testing 22

5.1 Libraries used 22

5.1.2 Implementation Procedure 23

5.5 Performance Metric Calculated 24

5.2 Code 27

5.3 Testing 34

5.4 Results 38

6. Conclusion and Future Scope 44

References 45

1. INTRODUCTION

In the era of the internet, the evaluation of user-generated opinions necessitates sophisticated automated systems. Current methods, which predominantly rely on reputation scores, often fall short in addressing the challenge of malicious manipulation. Recognizing this gap, our innovative approach takes a significant stride by integrating spam filtering and aspect-based sentiment analysis.

In contrast to prevailing systems that offer only entity-level scores, our project is designed to provide a more nuanced assessment through finer-grained aspect-based analysis. By leveraging cutting-edge deep learning algorithms and advanced natural language processing techniques, our objective is to elevate the accuracy and reliability of evaluating product and service reputations across a myriad of online platforms.

This novel methodology acknowledges the importance of not only gauging overall entity reputation but also understanding the specific aspects that contribute to the overall sentiment. By doing so, our approach not only enhances the depth of analysis but also acts as a robust defense against deceptive practices, ensuring a more authentic reflection of user sentiments and significantly improving the effectiveness of reputation assessments in the dynamic landscape of online platforms.

1.1 Motivation

In an era dominated by online interactions, the need to decipher authentic user sentiments from a sea of opinions is paramount. Traditional reputation scoring systems often overlook the intricate nuances of user-generated content, leaving room for malicious manipulations. Our project is motivated by a commitment to revolutionize this landscape. By fusing spam filtering and aspect-based sentiment analysis, we aspire to create a resilient framework that transcends the limitations of existing methods.

The motivation stems from the desire to provide businesses, consumers, and decision-makers with a robust, trustworthy tool for evaluating product and service reputations. Through advanced deep learning algorithms and natural language processing, we aim to empower users with fine-grained insights, fostering a more informed and discerning online community. This endeavor is driven by the belief that accurate, aspect-specific assessments are the cornerstone of building a reliable digital environment where authentic user opinions shine through, free from the distortions of deceptive practices.

1.2 Problem Statement

Existing online reputation systems lack the sophistication to combat malicious manipulations, often relying on oversimplified entity-level scores.

This project addresses the pressing problem of inadequate sentiment analysis by proposing an innovative solution that integrates spam filtering and aspect-based evaluation. The challenge lies in creating a more nuanced and reliable framework for assessing product and service reputations across diverse online platforms.

By leveraging deep learning and natural language processing, the goal is to overcome the limitations of current methods, offering users a trustworthy tool to navigate through the complexity of user-generated content, ultimately fostering a more authentic and informed digital landscape.

1.3 Project Objectives

The project aims to pioneer a holistic approach to online reputation management by integrating spam filtering and aspect-based sentiment analysis.

The primary objectives include developing advanced deep learning algorithms for accurate sentiment assessment, implementing robust spam detection mechanisms to counter manipulation, and providing nuanced, aspect-specific evaluations of product and service reputations.

By achieving these objectives, the project seeks to elevate the precision and reliability of online reputation assessments, offering a comprehensive solution that goes beyond conventional entity-level scoring and better equips users, businesses, and decision-makers with trustworthy insights in

the dynamic landscape of diverse online platforms.

1.4 Project Report Organization

This document contains six chapters. The first chapter contains motivation, problem statement, and objectives of the project. The second chapter includes existing works and limitations of existing works. The third chapter includes requirements and specifications.

The fourth chapter includes proposed solutions, UML diagrams and Technology Description. The fifth chapter includes implementation and testing. It also includes screenshots of the applications. The sixth chapter talks about the future enhancements and conclusion of the project.

2. LITERATURE SURVEY

Cross Platform Reputation Assessment from [1] reveals a growing need to automatically process the influx of opinions and reviews on Internet-based platforms. Existing systems, over the past decade, have focused on overall reputation generation from textual and numerical reviews, neglecting the impact of malicious user intentions. Addressing this gap, our proposed system integrates spam filtering, review popularity, posting time, and aspect-based sentiment analysis to provide accurate and reliable reputation values. Unlike previous models, it computes numerical reputations for both entities and their specific aspects. Experimental results across multiple platforms (Twitter, Facebook, Amazon) underscore the efficacy of our system compared to contemporary reputation generation systems.

Aspect-Based Sentiment Analysis for Reviews from [2] highlights the significance of aspect-based sentiment analysis (ABSA) involving aspect and sentiment extraction and sentiment orientation determination. Addressing this, our proposed system introduces a comprehensive approach to extract aspect sentiment pairs and calculate ratings for grouped aspects. The methodology encompasses subjective sentence selection, aspect and opinion extraction, sentiment orientation determination, and WordNet-based aspect clustering. Evaluation using precision, recall, and F1-Measure indicates strong performance, with F1-Measure values of 0.783 and 0.794 for subjectivity classification and extraction models, respectively. This research contributes to the advancement of ABSA methodologies, emphasizing precision and effectiveness in aspect-based sentiment analysis.

Addressing Spam in Reputation Evaluation from [3] underscores the intricacies of spam and spam filters within a complex social ecosystem. Traditional spam filtering systems often overlook the distributed and bulk nature of spam. Addressing this, the paper proposes a reputation-based collaborative anti-spam approach utilizing fingerprinting and assessing reporters' trust. Through comparison experiments on well-known email corpora, the

proposed approach demonstrates superior performance and robustness compared to state-of the-art techniques. This research contributes to the advancement of anti-spam methodologies, emphasizing the importance of collaborative reputation-based strategies for more effective spam filtering in diverse and dynamic online.

On the effectiveness of IP reputation for spam filtering from [4] highlights the significance of SMTP server mechanisms in mitigating spam, classifying them into pre-acceptance and post-acceptance approaches. Focusing on pre-acceptance filtering, particularly IP reputation, the paper categorizes SMTP senders into legitimate servers, end-hosts, and spam gangs. It empirically studies the effectiveness limits for each category and proposes innovative techniques to enhance IP reputation lists' performance. Leveraging DNS Sender Policy Framework (SPF), the paper develops custom lists and evaluates their effectiveness over time. The findings underscore the possibility of constructing highly accurate IP reputation lists, with the need for constant updates, offering valuable insights into mitigating spam across diverse sender types.

2.1 Existing work

Current approaches in reputation management predominantly rely on entity-level scores, neglecting nuanced aspects and susceptibility to spam. These methods lack the sophistication required to effectively process reviews from diverse platforms.

While some integrate sentiment analysis, their scope is limited, and they often overlook malicious manipulations. Advanced visualization tools are notably absent, hindering comprehensive data interpretation.

The existing work highlights a gap in handling the intricacies of online reputation assessment, paving the way for our project to introduce a novel framework.

By combining spam filtering, aspect-based sentiment analysis, and advanced visualization, we aim to address these deficiencies, providing a more comprehensive and reliable solution.

2.1 Limitations of Existing work

i. **Limited Source Integration:**

- Current systems struggle to effectively process reviews from a diverse range of platforms, leading to incomplete and biased assessments.

ii. **Spam Vulnerability:**

- The existing approach lacks robust spam filtering mechanisms, making it susceptible to manipulations that can distort the accuracy of reputation evaluations.

iii. **Lack of Nuanced Assessments:**

- Entity-level scoring in the current system overlooks specific aspects or attributes, such as product quality or customer service, resulting in a lack of nuanced evaluations

iv. **Absence of Advanced Visualization:**

- The current framework lacks sophisticated visualization tools, hindering the interpretation of complex data patterns and trends in reputation dynamics.

v. **Inability to Adapt to Diverse Platforms:**

- The limitations include a failure to adapt to the dynamic nature of various online platforms, impeding the system's ability to provide comprehensive and reliable assessments.

vi. **Scalability Challenges:**

Current models may struggle with scalability issues when dealing with large datasets, impacting the efficiency and speed of reputation assessments.

3. SOFTWARE AND HARDWARE REQUIREMENTS

Operating system	:	windows only
Processor	:	i5 and above
Ram	:	8gb and above
Hard disk	:	25gb and above
Software	:	Anaconda 3

3.1 Functional Requirements:

The project requires robust functionality to address the limitations of existing systems in evaluating product, service, or entity reputations across diverse online platforms. The software should be able to perform the following functions:

- Data collection
- Spam detection
- Aspect-Based Sentimental Analysis
- Multisource Integration
- Advanced Visualization Tools
- Scalability

3.2 Non-Functional Requirements:

In the realm of software development for the proposed project, key non-functional requirements play a pivotal role in ensuring the system's overall effectiveness, performance, and user satisfaction. The non-functional requirements for this software are as follows:

- Response time
- Throughput

- Scalability
- Reliability and Availability
- Security
- Interoperability
- Maintainability
- Usability
- Adaptability
- Compliance
- Performance Monitoring and Logging

4.SYSTEM ANALYSIS

4.1 MODULES:

The system is divided into 5 different modules:

- Data Collection Module
- Spam Filtering module
- Aspect Based sentimental Analysis module
- Numerical reputation value computational module
- Visualization Tool module

- **Data Collection module**:

  The Data Collection module systematically collects reviews and opinions from diverse platforms, including Twitter, Facebook, Amazon, and others. By interfacing with APIs, web scraping tools, or other relevant methods, this module aggregates textual data from multiple sources. The collected information becomes the foundation for subsequent analysis, allowing the system to harness a broad spectrum of user-generated content for a comprehensive understanding of sentiments and opinions related to the targeted entities, enabling a nuanced approach to aspect-based sentiment analysis.

- **Spam Filtering module**:

  The Spam Filtering module is dedicated to discerning and excluding reviews originating from malicious users or spam accounts. Employing advanced algorithms, it analyzes patterns and behaviors to identify deceptive content. This module enhances the reliability of sentiment analysis by systematically eliminating fraudulent or misleading opinions, ensuring that the evaluation of product or service reputation remains robust and untainted by spam. Through its proactive filtering mechanisms, the Spam Filtering module contributes to the production of more accurate and trustworthy sentiment assessments across diverse online platforms.

- **Aspect-Based Sentiment Analysis module**:

  The Aspect-Based Sentiment Analysis module meticulously evaluates sentiment for individual aspects, offering a nuanced comprehension of user opinions. Employing advanced sentiment analysis techniques, it discerns whether each aspect is perceived positively or negatively. This granular approach enables a more detailed examination of user sentiments, distinguishing between various facets of opinions and providing valuable insights into the specific aspects that contribute to overall sentiment, enhancing the precision and depth of sentiment analysis.

- **Numerical Reputation Value Computation module**:

  The Numerical Reputation Value Computation module integrates outcomes from preceding modules to calculate numerical reputation values for both the overall entity and its distinct aspects. By consolidating data gathered through sentiment analysis, spam detection, and other relevant processes, this module provides a quantitative measure of reputation. It ensures a comprehensive evaluation, offering a nuanced and detailed understanding of the entity's reputation, fostering more precise and reliable assessments across various dimensions and aspects.

- **Visualization Tool module**:

  The Visualization Tool module is dedicated to crafting a sophisticated interface that presents comprehensive details about the generated reputation values. This tool employs advanced visualization techniques to offer a user-friendly display, enabling stakeholders to easily interpret and understand the intricacies of reputation assessments. By providing detailed insights into the strengths and weaknesses of entities, the module enhances the user experience, fostering informed decision-making. The visualizations serve as a key

component in conveying nuanced aspects of reputation, contributing to a more transparent and insightful representation of the analyzed data.

4.2 ALGORITHMS USED

Algorithm used in this project are Regression algorithms such as Navie Bayes Algorithm, Logistic Algorithm, SVM Algorithm and K-nearest Algorithm.

4.2.1 Navie Bayes Algorithm:

The Naive Bayes algorithm is a probabilistic classification method based on Bayes' theorem, assuming independence among features. It's widely used for text classification and spam filtering. The formula for Naive Bayes is:

$$P(C|X)= P(X|C)\cdot P(C)/P(X)$$

Where:

- $P(C|X)$ is the posterior probability of class C given feature set X.
- $P(X|C)$ is the likelihood, the probability of observing X given C.
- $P(C)$ is the prior probability of class C.
- $P(X)$ is the probability of observing X.

Naive Bayes simplifies computations by assuming independence between features, making it computationally efficient and effective for various classification tasks.

Naive Bayes classifiers are known for their simplicity and effectiveness in text classification tasks, including spam detection. They can be employed to assess whether a particular review or opinion is likely to be spam or genuine, contributing to the overall accuracy of sentiment analysis and reputation assessments in the project.

4.2.2 Logistic Algorithm

Logistic Regression is a binary classification algorithm used to predict the probability of an instance belonging to a particular class. It models the relationship between a dependent binary variable and one or more independent variables. The logistic function (sigmoid) is central to this algorithm, and the probability is computed using the formula:

P(Y=1)= 1/ 1+e ^−(b0+b1.x1+b2.x2 +...+bn.xn)

Here,

- P(Y=1) is the probability of the positive class,
- e is the base of the natural logarithm,
- b0 is the intercept, b1,b2,…,bn are the coefficients, and
- x1,x2,….,xn are the independent variables.

Logistic Regression uses techniques like Maximum Likelihood Estimation to optimize these coefficients during the training process, producing a model capable of predicting probabilities and making binary classifications.

Logistic Regression in this project could be employed as the classifier in the sentiment analysis model. By utilizing the probabilities predicted by Logistic Regression, the project can calculate numerical reputation values for entities and their specific aspects, providing a quantitative measure of reputation, contributing to a more nuanced understanding of online sentiment.

4.2.3 SVM Algorithm

Support Vector Machine (SVM) is a powerful supervised machine learning algorithm used for classification and regression tasks. It works by finding the optimal hyperplane that separates data into different classes. The algorithm aims to maximize the margin between classes while minimizing classification errors. The hyperplane is determined by support vectors, which are data points closest to the decision boundary.

The formula for the hyperplane is expressed as:

$$w \cdot x + b = 0$$

where:

- w is the weight vector perpendicular to the hyperplane,
- x represents the input features,
- b is the bias term.

The decision function is $f(x)=w \cdot x+b$, and the predicted class is determined by the sign of $f(x)$. SVM is effective in high-dimensional spaces and is versatile for various applications.

In the above project, SVM may be employed for sentiment analysis and spam detection within the aspect-based reputation system. SVM helps classify reviews into positive or negative sentiments, contributing to the numerical reputation values calculation. Additionally, it aids in spam detection, enhancing the accuracy of sentiment analysis for more reliable online reputation assessments.

4.2.4 K-nearest Algorithm.

The K-Nearest Neighbors (K-NN) algorithm is a simple and intuitive supervised machine learning approach used for classification and regression tasks. Given a dataset, it classifies a new data point based on the majority class of its k nearest neighbors. The algorithm relies on the assumption that similar data points in the feature space will share the same class or have similar regression values. The classification or regression decision is made by a majority vote or averaging the values of the k nearest neighbors.

Formula for Euclidean Distance (for two-dimensional space):

Distance=root $(x_2-x_1)^2+(y_2-y_1)^2$

Here,x1,y1 are the coordinates of one data point, and x2,y2 are the coordinates of another data point. The algorithm calculates the distance between these points and selects the k nearest neighbors for classification or regression.

The K-Nearest Neighbors (K-NN) algorithm can be utilized for various tasks such as sentiment analysis or spam detection. By measuring the similarity between data points in the aspect-based

sentiment analysis, K-NN helps assess reputation values by considering the opinions of neighboring data points, contributing to a more nuanced evaluation.

4. SYSTEM DESIGN

The "Multiplatform Reputation Analysis System" addresses the shortcomings of existing models to furnish consumers with trustworthy and detailed reputation information. In response to the limitations of current systems, our approach involves a comprehensive evaluation of user opinions spanning various platforms.

Unlike conventional methods, our system emphasizes specific aspects, offering a nuanced analysis that goes beyond generic entity-level scoring. This precision in assessment ensures that consumers receive highly accurate insights into the strengths and weaknesses of products and services.

By facilitating a more refined and platform-diverse reputation management process, our system empowers users to make well-informed decisions. It aligns with the evolving dynamics of online discourse, providing a reliable solution for those seeking robust, multiplatform reputation insights that cater to the intricacies of diverse consumer preferences and expectations.

4.1 Proposed Approaches

The "Multiplatform Reputation Analysis System" employs a multifaceted approach to enhance the precision and reliability of reputation information. The proposed methods include:

Cross-Platform Data Aggregation; Develop algorithms to efficiently aggregate and normalize user-generated content from diverse online platforms, ensuring a comprehensive dataset for analysis.

Spam Detection Mechanisms: Integrate robust spam filtering algorithms to identify and exclude fraudulent or manipulated reviews, enhancing the accuracy and authenticity of reputation assessments.

Aspect-Based Sentiment Analysis: Implement advanced natural language processing and sentiment analysis techniques to dissect user opinions and evaluate reputation based on specific aspects such as product quality, customer service, and other relevant attributes.

User-Driven Weighting System: Introduce a user-driven weighting system, allowing consumers to prioritize specific aspects based on their preferences, tailoring the reputation analysis to individual user needs.

Visualization Tools: Develop intuitive and advanced visualization tools to present reputation insights in a user-friendly manner, aiding in the interpretation of complex data patterns and trends.

Continuous Learning and Updates: Establish mechanisms for continuous learning, allowing the system to adapt and improve over time through feedback loops and updates based on evolving user sentiments and online trends.

4.2 UML DIAGRAMS

UML stands for Unified Modelling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group. The goal is for UML to become a common language for creating models of object-oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML. The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems. The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems. The UML is a very important part of developing objects-oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.
4. Provide a formal basis for understanding the modelling language.
5. Support higher level development concepts such as collaborations, frameworks,

patterns and components

4.2.1 Use Case Diagram

Figure 4.2.1 depicts the Use Case Diagram for the Multiplatform Reputation Analysis System provides a visual representation of the system's functionalities. It portrays actors, namely the User, and the Model, interacting with key use cases.

The process begins with the User initiating the collection of opinions and reviews, which are then processed. Subsequently, the system incorporates spam filtering techniques to enhance data accuracy. The pivotal "Generate Reputation System" use case encapsulates the core functionality, producing numerical reputation values for entities and their aspects.

The "Visual Representation" use case involves the advanced visualization tool, displaying detailed information about generated reputation values.

The final use case, "Results," encompasses the delivery of comprehensive reputation insights to the User. This diagram not only outlines the functionalities performed by the system but also clarifies the roles and interactions of actors, offering a holistic overview of the Multiplatform Reputation Analysis System's operation.

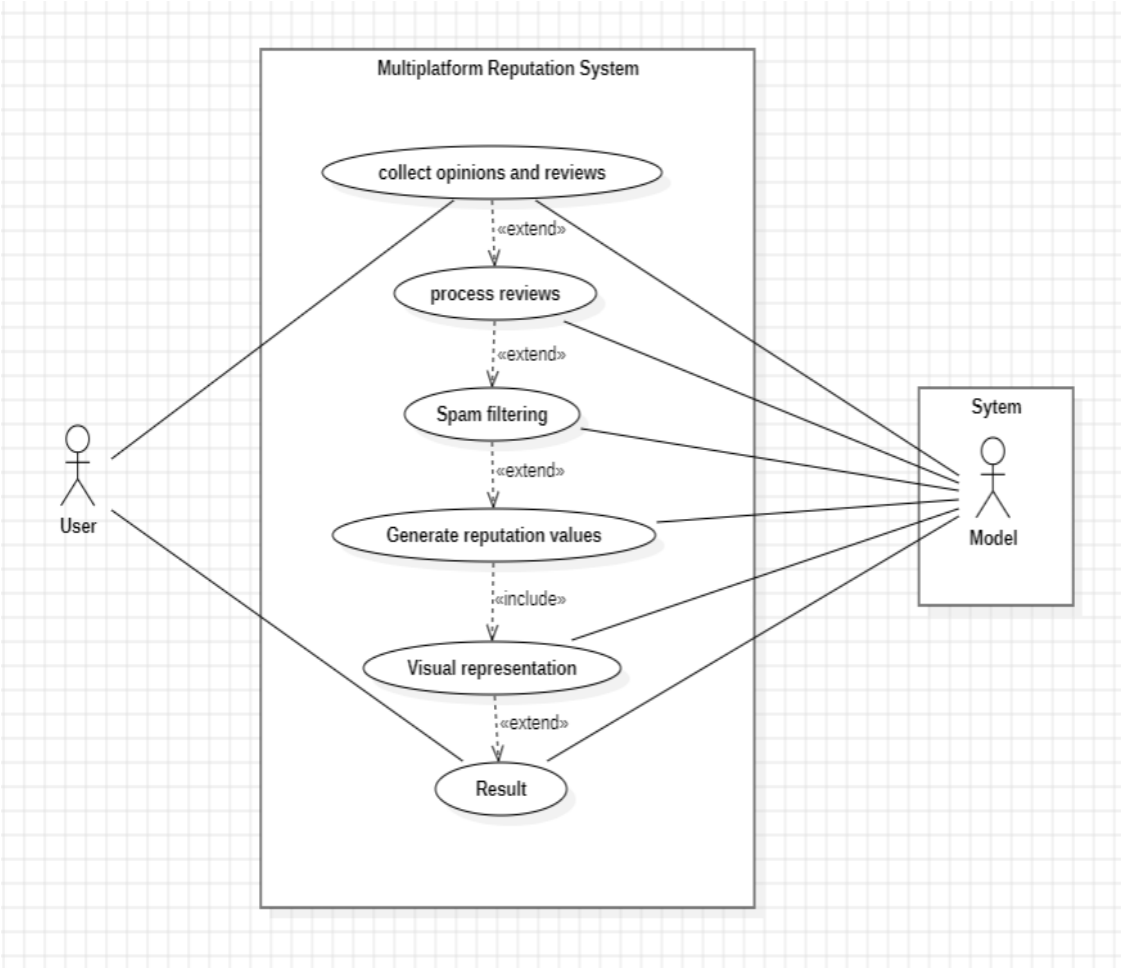

Fig.4.2.1. Use Case Diagram for Multiplatform Reputation Analysis System

4.2.2 Class Diagram

Figure 4.2.2 depicts the Class Diagram for the Multiplatform Reputation Analysis System illustrates the key classes involved.

These include "User," representing system users, "Multiplatform Reputation System," the core module orchestrating reputation analysis, "Reputation Builder," responsible for calculating reputation values, "Opinion Collector," gathering data from diverse sources, and "Sentiment," handling sentiment analysis.

This visual representation delineates relationships and interactions among these classes, showcasing attributes and operations/methods associated with each. It serves as a blueprint for developers, facilitating a clear understanding of the system's architecture, aiding in the design, implementation, and maintenance phases of software development.

The diagram acts as a visual guide, enhancing communication among development teams and stakeholders about the structure and behavior of the Multiplatform Reputation Analysis System.

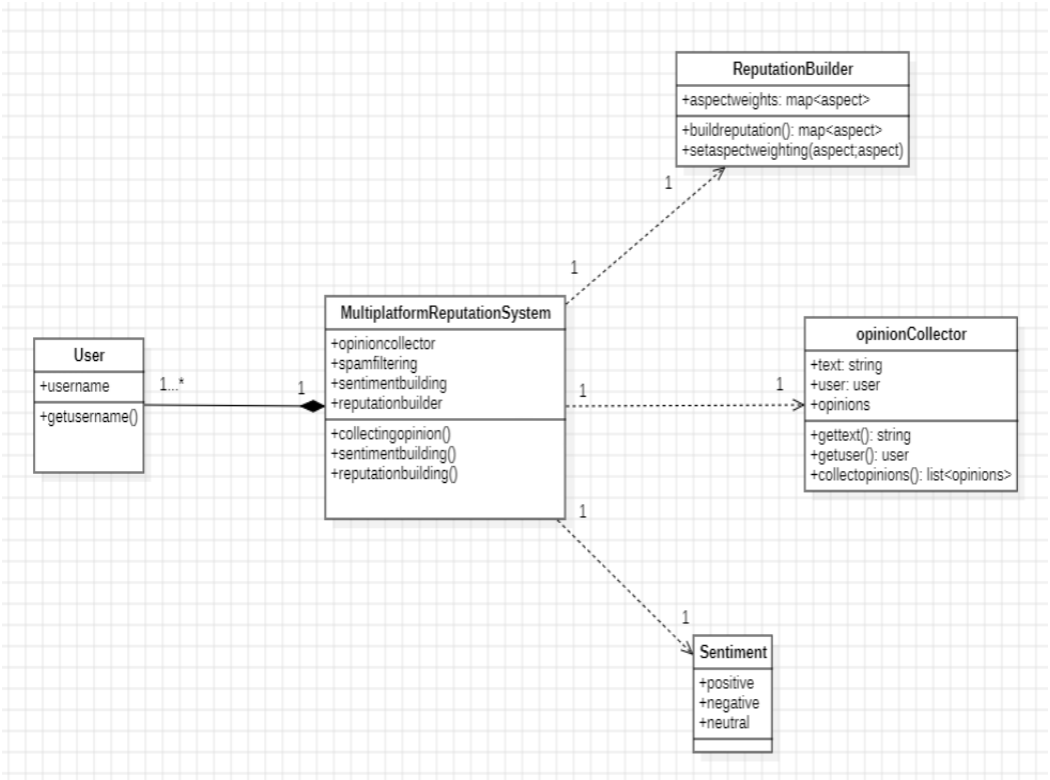

Fig.4.2.2. Class Diagram for Multiplatform Reputation Analysis System

4.2.3 Sequence Diagram

A Unified Modeling Language (UML) Sequence Diagram visually represents interactions and order of processes in a system. It's akin to a Message Sequence Chart, also known as event diagrams, event scenarios, or timing diagrams.

Fig.4.2.3 illustrates the Sequence Diagram for the Multiplatform Reputation Analysis System. The diagram showcases classes such as User, Multiplatform Reputation System, Data Collection, Spam Filtering, Reputation Calculation, and Visualization Tool. Users initiate the process by providing opinions and reviews, triggering a sequence of interactions.

These interactions involve data collection, spam filtering, reputation calculation, and the eventual visualization of results.

The diagram effectively communicates the dynamic flow of processes, providing a clear representation of how various components collaborate in the Multiplatform Reputation Analysis System.

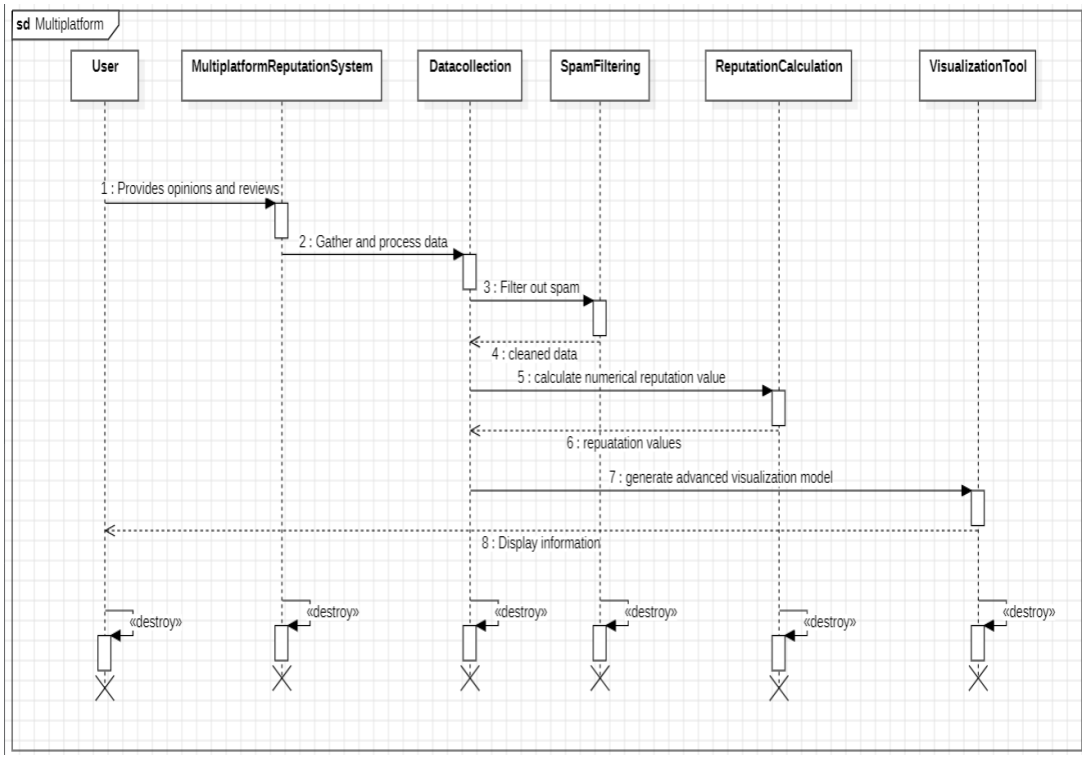

Fig.4.2.3. Sequence Diagram for Multiplatform Reputation Analysis System

4.2.4 Activity Diagram

Figure 4.2.4 illustrates the Activity Diagram for the Multiplatform Reputation Analysis System illustrates the sequential workflow of key actions.

It begins with the generation of opinions and reviews, progresses to the processing of reviews, and incorporates spam filtering for enhanced accuracy. Subsequently, reputation values are generated, reflecting a numerical assessment of entity reputation.

The final step involves the display of detailed information, providing users with comprehensive insights. This structured representation allows stakeholders to visualize the stepwise execution of critical processes within the system.

The diagram's use of Unified Modeling Language ensures clarity in describing the business and operational workflows, facilitating effective communication and understanding among project stakeholders.

Overall, the Activity Diagram serves as a valuable tool for system design, depicting the orchestrated sequence of actions contributing to the Multiplatform Reputation Analysis System's functionality and performance.

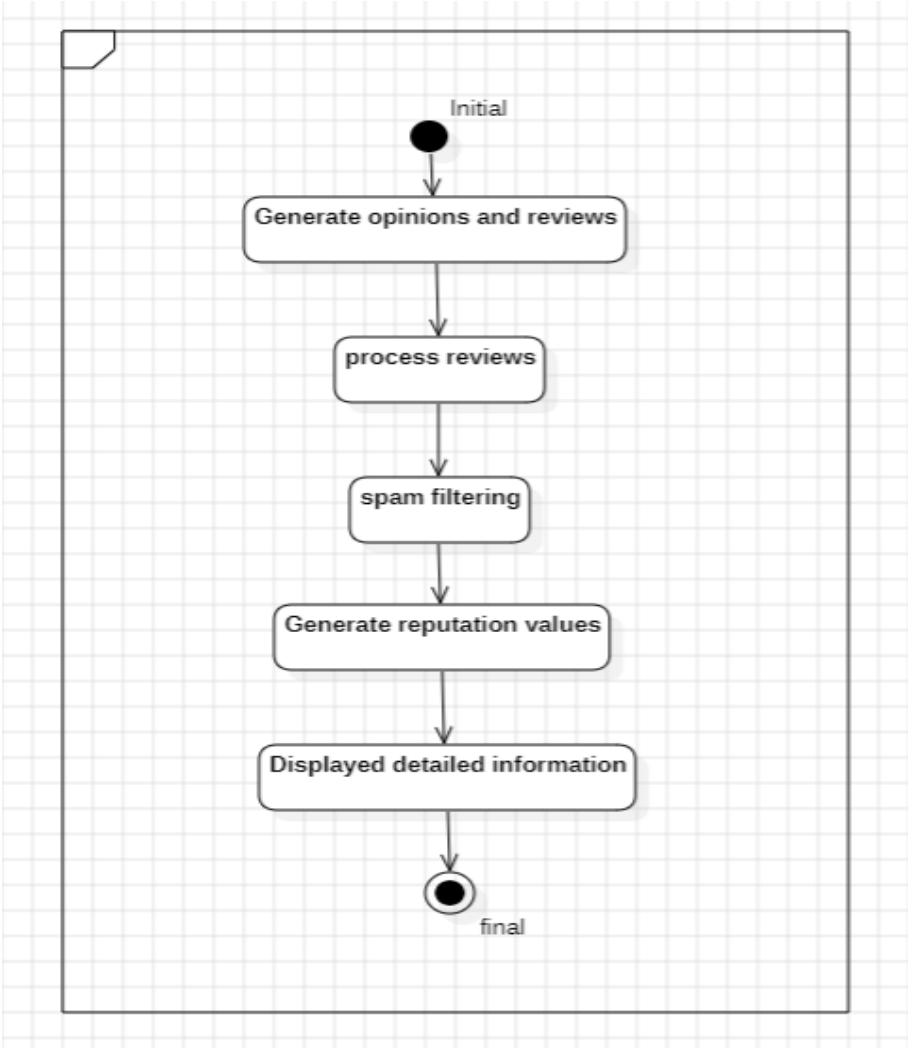

Fig.4.2.4. Activity Diagram for Multiplatform Reputation Analysis System

5. IMPLEMENTATION AND TESTING

5.1 Libraries used

The necessary libraries are pandas, numpy ,matplotlib.pyplot, seaborn

Pandas: Pandas is a powerful and popular data manipulation library for Python. It provides data structures like DataFrame and Series, which are efficient for working with structured data.

NumPy: NumPy, short for Numerical Python, is a fundamental package for scientific computing in Python. It provides support for large, multi-dimensional arrays and matrices, along with mathematical functions to operate on these arrays.

Matplotlib.pyplot: Matplotlib is a 2D plotting library for Python that produces high-quality figures. pyplot is a module within Matplotlib that provides a MATLAB-like interface for creating a variety of plots and charts.

Seaborn: Seaborn is a statistical data visualization library built on top of Matplotlib. It provides a high-level interface for creating informative and attractive statistical graphics.

5.1.1 Implementation Procedure:

To implement the specified tasks, first, ensure Anaconda is installed on your system. Open a Jupyter Notebook and import essential libraries like pandas, numpy, matplotlib.pyplot, and seaborn.

After installing any missing packages, load the 'yelp.csv' dataset into a pandas DataFrame, displaying initial rows to explore the data. Calculate the 'text length' for each review, adding a new column to the DataFrame.

Generate visualizations, including histograms and box plots, to examine the distribution of text lengths across distinct star ratings, gaining insights into potential patterns. Split the dataset into training and testing sets.

Create pipelines for text vectorization using CountVectorizer and classification using Naive Bayes (MultinomialNB), Logistic Regression, SVM, and K-Nearest Neighbors algorithms.

Train the pipelines on the training data and make predictions on the test set. Evaluate model performance using classification reports and confusion matrices, providing insights into precision, recall, and accuracy.

Conclude results based on the accuracy achieved by each algorithm, allowing a comparison of their effectiveness in the context of the Yelp dataset.

This comprehensive approach ensures a thorough exploration of the data, robust model training, and insightful evaluation, enabling informed decisions regarding the most suitable algorithm for the given text classification task.

5.1.2 PERFORMANCE METRICS CALCULATED

Precision: Precision is the ratio of true positive predictions to the total predicted positives. It assesses the accuracy of positive predictions.

Precision=True Positives / True Positives + False Positives

Recall (Sensitivity or True Positive Rate): Recall is the ratio of true positives to the total actual positives, evaluating prediction completeness.

Recall= True Positives / True Positives + False Negatives

F1-Score: F1-Score is the harmonic mean of precision and recall, providing a balanced measure.

F1-score =2 ×Precision × Recall / Precision + Recall

Support: Support is the number of occurrences of each class in the actual data. It provides insights into the distribution of different sentiment classes or aspects, aiding in understanding the prevalence of specific categories and contributing to a more comprehensive evaluation of the sentiment analysis results.

Confusion Matrix: A confusion matrix is a matrix that summarizes the performance of a machine learning model on a set of test data. It is often used to measure the performance of classification models, which aim to predict a categorical label for each input instance. The matrix displays the number of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) produced by the model on the test data.

	Actual	
Predicted	TP	FP
	FN	TN

Accuracy: Accuracy is the ratio of correct predictions to the total predictions, measuring overall correctness.

Accuracy= Total Predictions / True Positives + True Negatives

Macro Average: Macro Average is employed to calculate the unweighted average of precision, recall, and F1-Score across all classes. It ensures equal contribution from each class, providing a holistic performance measure that isn't influenced by imbalances in class sizes, offering a comprehensive evaluation of the model's effectiveness.

Weighted Average: Weighted Average is crucial for assessing performance metrics like precision, recall, and F1-Score. It accounts for class imbalances by giving more importance to metrics in classes with larger datasets, ensuring a comprehensive evaluation that considers the impact of varying data distributions on the overall model performance.

5.2 CODE:

```
#importing data manipulation and visualization libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

#Loading Dataset
yelp=pd.read_csv('yelp.csv')

yelp.head()
yelp.describe()
import string

t1=yelp['text']
t1
#Creating a new column 'text length' which stores length of each review
yelp['text length']=yelp['text'].apply(len)

yelp
#Histogram to visualize text length in each star rating
g=sns.FacetGrid(yelp,col='stars')
g.map(plt.hist,'text length')
sns.boxplot(x='stars',y='text length',data=yelp)
#count of reviews in each rating
sns.countplot(x='stars',data=yelp)
```

```
yelp.groupby('stars').mean().corr()
#Heatmap to find if there is any correlation among any feature, most correlated features being funny and useful
sns.heatmap(yelp.groupby('stars').mean().corr(),cmap='coolwarm',annot=True)
#Since we are only classifying data as 1 or 5 star review, this step is used to separate that specific data
yelp_class=yelp[yelp['stars'].apply(lambda x: x==1 or x==5)]

#Features and Target variables
x=yelp_class['text']
y=yelp_class['stars']

#Importing libraries for Splitting, Vectorizing the data and also the algorithm to be used
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.naive_bayes import MultinomialNB

#Splitting the data as training and testing data sets, test size set to 30 percent off the main data
x=yelp_class['text']
y=yelp_class['stars']
X_train, X_test, y_train, y_test = train_test_split(x, y, test_size=0.3, random_state=101)

#Creating a pipeline to make execution simpler and using Naive Bayes as Classifying algorithm
pipeline=Pipeline([('Count Vectorizer',CountVectorizer()),
                   ('Classifier',MultinomialNB())
])
pipeline.fit(X_train,y_train)

Pipeline(memory=None,
         steps=[('Count Vectorizer',
```

```
            CountVectorizer(analyzer='word', binary=False,
                            decode_error='strict',
                            dtype=<class 'numpy.int64'>, encoding='utf-8',
                            input='content', lowercase=True, max_df=1.0,
                            max_features=None, min_df=1,
                            ngram_range=(1, 1), preprocessor=None,
                            stop_words=None, strip_accents=None,
                            token_pattern='(?u)\\b\\w\\w+\\b',
                            tokenizer=None, vocabulary=None)),
           ('Classifier',
            MultinomialNB(alpha=1.0, class_prior=None, fit_prior=True))],
         verbose=False)

predict=pipeline.predict(X_test)

#Observations
from sklearn.metrics import classification_report,confusion_matrix
print(classification_report(predict,y_test))
print(confusion_matrix(predict,y_test))
yelp_class.head()
#Plotting Stars vs Text Length
sns.jointplot(x='text length',y='stars',data=yelp_class)
#Using a Logistic Regression Model
from sklearn.linear_model import LogisticRegression
pipeline2=Pipeline([('Count Vectorizer',CountVectorizer()),
                    ('Classifier',LogisticRegression())
])

pipeline2.fit(X_train,y_train)
Pipeline(memory=None,
         steps=[('Count Vectorizer',
```

```
            CountVectorizer(analyzer='word', binary=False,
                            decode_error='strict',
                            dtype=<class 'numpy.int64'>, encoding='utf-8',
                            input='content', lowercase=True, max_df=1.0,
                            max_features=None, min_df=1,
                            ngram_range=(1, 1), preprocessor=None,
                            stop_words=None, strip_accents=None,
                            token_pattern='(?u)\\b\\w\\w+\\b',
                            tokenizer=None, vocabulary=None)),
           ('Classifier',
            LogisticRegression(C=1.0, class_weight=None, dual=False,
                               fit_intercept=True, intercept_scaling=1,
                               l1_ratio=None, max_iter=100,
                               multi_class='auto', n_jobs=None,
                               penalty='l2', random_state=None,
                               solver='lbfgs', tol=0.0001, verbose=0,
                               warm_start=False))],
         verbose=False)

predict2=pipeline2.predict(X_test)

print(classification_report(predict2,y_test))
print(confusion_matrix(predict2,y_test))
from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score

from sklearn.model_selection import train_test_split

from sklearn.feature_extraction.text import CountVectorizer

from sklearn.pipeline import Pipeline
```

```
# Assuming X_train, X_test, y_train, y_test are your training and testing data

# You might need to replace this with your actual data loading and preprocessing code.


# Split the data into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(features, labels, test_size=0.2,
random_state=42)


# Create a pipeline with CountVectorizer and LinearRegression

pipeline_linear = Pipeline([

    ('Count Vectorizer', CountVectorizer()),

    ('Regressor', LinearRegression())

])


# Fit the pipeline on the training data

pipeline_linear.fit(X_train, y_train)


# Make predictions on the test data

predict_linear = pipeline_linear.predict(X_test)


# Evaluate the performance of the Linear Regression model

mse = mean_squared_error(y_test, predict_linear)
```

```
r2 = r2_score(y_test, predict_linear)


# Print the evaluation metrics
print(f'Mean Squared Error: {mse}')
print(f'R-squared Score: {r2}')
from sklearn.svm import SVC
from sklearn.pipeline import Pipeline
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics import classification_report, confusion_matrix


# Assuming X_train, y_train, X_test, y_test are your training and testing data
# If not, replace them with your actual data


# Creating a pipeline with CountVectorizer and SVM Classifier
pipeline_svm = Pipeline([
    ('Count Vectorizer', CountVectorizer()),
    ('Classifier', SVC())  # Using Support Vector Machine Classifier
])


# Fitting the pipeline on the training data
pipeline_svm.fit(X_train, y_train)
```

```
# Making predictions on the test data

predict_svm = pipeline_svm.predict(X_test)


# Printing classification report and confusion matrix

print("Classification Report:")

print(classification_report(predict_svm, y_test))


print("\nConfusion Matrix:")

print(confusion_matrix(predict_svm, y_test))
```

5.3 TESTING

5.3.1 TestCase-1: Uploading Dataset

Dataset **yelp.csv** will be uploaded using yelp=pd**.**read_csv('yelp.csv')

	business_id	date	review_id	stars	text	type	user_id	cool	useful	funny
0	9yKzy9PApeiPPOUJEtnvkg	2011-01-26	fWKvX83p0-ka4JS3dc6E5A	5	My wife took me here on my birthday for breakf...	review	rLtl8ZkDX5vH5nAx9C3q5Q	2	5	0
1	ZRJwVLyzEJq1VAihDhYiow	2011-07-27	IjZ33sJrzXqU-0X6U8NwyA	5	I have no idea why some people give bad review...	review	0a2KyEL0d3Yb1V6aivbIuQ	0	0	0
2	6oRAC4uyJCsJl1X0WZpVSA	2012-06-14	lESLBzqUCLdSzSqm0eCSxQ	4	love the gyro plate. Rice is so good and I als...	review	0hT2KtfLiobPvh6cDC8JQg	0	1	0
3	_1QQZuf4zZOyFCvXc0o6Vg	2010-05-27	G-WvGaISbqqaMHlNnByodA	5	Rosie, Dakota, and I LOVE Chaparral Dog Park!!...	review	uZetl9T0NcROGOyFfughhg	1	2	0
4	6ozycU1RpktNG2-1BroVtw	2012-01-05	1uJFq2r5QfJG_6ExMRCaGw	5	General Manager Scott Petello is a good egg!!!...	review	vYmM4KTsC8ZfQBg-j5MWkw	0	0	0

Fig.5.3.1 Testing to upload Dataset

Remark: Successfully Uploaded

5.3.2 Test Case-2: Creating a new column text length

Text length will be created using yelp['text length']=yelp['text']**.**apply(len)

	business_id	date	review_id	stars	text	type	user_id	cool
0	9yKzy9PApeiPPOUJEtnvkg	2011-01-26	fWKvX83p0-ka4JS3dc6E5A	5	My wife took me here on my birthday for breakf...	review	rLtl8ZkDX5vH5nAx9C3q5Q	2
1	ZRJwVLyzEJq1VAihDhYiow	2011-07-27	lJZ33sJrzXqU-0X6U8NwyA	5	I have no idea why some people give bad review...	review	0a2KyEL0d3Yb1V6aivbIuQ	0
2	6oRAC4uyJCsJl1X0WZpVSA	2012-06-14	IESLBzqUCLdSzSqm0eCSxQ	4	love the gyro plate. Rice is so good and I als...	review	0hT2KtfLiobPvh6cDC8JQg	0
3	_1QQZuf4zZOyFCvXc0o6Vg	2010-05-27	G-WvGaISbqqaMHlNnByodA	5	Rosie, Dakota, and I LOVE Chaparral Dog Park!!...	review	uZetl9T0NcROGOyFfughhg	1
4	6ozycU1RpktNG2-1BroVtw	2012-01-05	1uJFq2r5QfJG_6ExMRCaGw	5	General Manager Scott Petello is a good egg!!!...	review	vYmM4KTsC8ZfQBg-j5MWkw	0
...	...	...	...	...	...	...	...	...
9995	VY_tvNUCCXGXQeSvJl757Q	2012-07-28	Ubyfp2RSDYW0g7Mbr8N3iA	3	First visit...Had lunch here today - used my G...	review	_eqQoPtQ3e3UxLE4faT6ow	1
9996	EKzMHI1tip8rC1-ZAy64yg	2012-01-18	2XyIOQKbVFb6uXQdJ0RzlQ	4	Should be called house of deliciousness!\n\n| ...	review	ROru4uk5SaYc3rg8IU7SQw	0

Fig. 5.3.2 Testing to create a text length column

Remark: Successfully Created

5.3.3 TestCase-3: Counting reviews in each rating

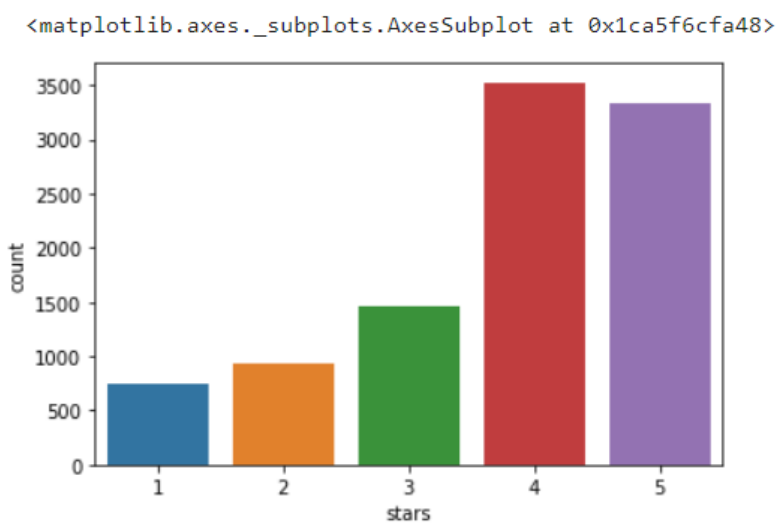

Fig. 5.3.3 Testing for review counts using stars

Remark: Successfully Done

5.3.4 TestCase-4: Checking any correlation between features

Finding any correlation between features using

sns.heatmap(yelp.groupby('stars').mean().corr(),cmap='coolwarm',annot=**True**)

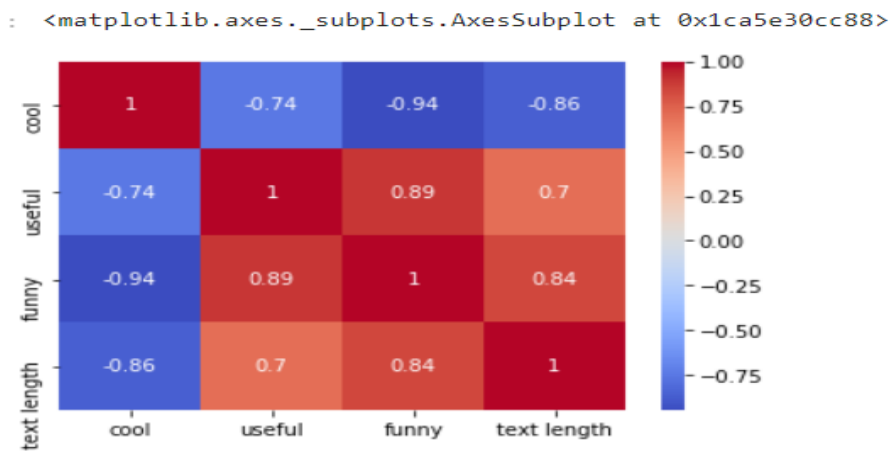

Fig.5.3.4 Testing for correlation between features

Remark: Successfully Tested

5.3.5 TestCase-5: Testing with Algorithms

Algorithms are tested by creating a pipeline to make execution simpler.

Remark: Successfully Tested

Test Case Id	Test Case Name	Test Description	Expected Output	Actual Output	Remark
1	Uploading Dataset	A data set of brain imaging features	Dataset should be uploaded .	Dataset is uploaded	SUCCESS
2	Creating a new column text length	To create a text length column	Text column should be created	Column is created	SUCCESS
3	Testing for correlation between features	To create a visualization by using graphs	Correlation Matrix should be obtained	Correlation Matrix is obtained	SUCCESS
4	Machine learning model training and evaluation	Training multiple classifiers and evaluating their performance metrics (accuracy, precision, recall, and F1 score).	Accurate values should be acquired for performance metrics	Values are acquired	SUCCESS
5	To draw a final conclusion	To get a final conclusion from the acquired values from training the algorithms	Multiplatform Reputation Analysis System should be obtained	Multiplatform Reputation Analysis System is obtained	SUCCESS

5.4 RESULTS

5.4.1 RESULT-1: When Navie Bayes Algorithm is used

OUTPUT:

```
              precision    recall  f1-score   support

           1       0.68      0.91      0.78       170
           5       0.98      0.93      0.96      1056

    accuracy                           0.93      1226
   macro avg       0.83      0.92      0.87      1226
weighted avg       0.94      0.93      0.93      1226

[[155  15]
 [ 73 983]]
```

Fig.5.4.1 Output with Navie Bayes Algorithm

Rating	Precision	Recall	F1-Score	Support
1	0.68	0.91	0.78	170
5	0.98	0.93	0.96	1056

Table.5.4.1 Output values for Navie Bayes Algorithm

Table.5.4.1 illustrates the precision,Recall,F1-score and Support values for the rating 1 and 5 where 1 is the lowest rating and 5 is the highest ratng for the Navie Bayes algorithm .

5.4.2 RESULT-2: When Logistic Regression algorithm is used

OUTPUT:

```
              precision    recall  f1-score   support

           1       0.77      0.87      0.82       201
           5       0.97      0.95      0.96      1025

    accuracy                           0.94      1226
   macro avg       0.87      0.91      0.89      1226
weighted avg       0.94      0.94      0.94      1226

[[175  26]
 [ 53 972]]
```

Fig.5.4.2 Output with Logistic Regression Algorithm

Rating	Precision	Recall	F1-Score	Support
1	0.77	0.87	0.82	201
5	0.97	0.95	0.96	1025

Table.5.4.2 Output values for Logistic Regression Algorithm

Table.5.4.2 illustrates the precision,Recall,F1-score and Support values for the rating 1 and 5 where 1 is the lowest rating and 5 is the highest ratng for the Logistic Regression Algorithm.

5.4.3 RESULT-3: When SVM Regression algorithm is used

OUTPUT:

```
Classification Report:
              precision    recall  f1-score   support

           1       0.36      0.98      0.53        84
           5       1.00      0.87      0.93      1142

    accuracy                           0.88      1226
   macro avg       0.68      0.92      0.73      1226
weighted avg       0.95      0.88      0.90      1226


Confusion Matrix:
[[ 82   2]
 [146 996]]
```

Fig.5.4.3 Output with SVM Regression Algorithm

Rating	Precision	Recall	F1-Score	Support
1	0.36	0.98	0.53	84
5	1.00	0.87	0.93	1142

Table.5.4.3 Output values for SVM Regression Algorithm

Table.5.4.3 illustrates the precision,Recall,F1-score and Support values for the rating 1 and 5 where 1 is the lowest rating and 5 is the highest ratng for the SVM Algorithm

5.4.4 RESULT-4: When K-nearest neighbour algorithm is used

OUTPUT:

```
              precision    recall  f1-score   support

           1       0.09      0.72      0.16        29
           5       0.99      0.83      0.90      1197

    accuracy                           0.82      1226
   macro avg       0.54      0.78      0.53      1226
weighted avg       0.97      0.82      0.88      1226

[[ 21   8]
 [207 990]]
```

Fig.5.4.4 Output with K-nearest Algorithm

Rating	Precision	Recall	F1-Score	Support
1	0.09	0.72	0.16	29
5	0.99	0.83	0.90	1197

Table.5.4.4 Output values for K-nearest Algorithm

Table.5.4.4 illustrates the precision, Recall, F1-score and Support values for the rating 1 and 5 where 1 is the lowest rating and 5 is the highest ratng for the K-Nearest Neighbor Algorithm.

Comparative Analysis of Algorithms:

All the models gave accuracy with Naive Bayes at 93% , Logistic Regression at 94% , SVM Algorithm at 88% and K-nearest Algorithm at 82%

Though Logistic Regression has overall better precision.

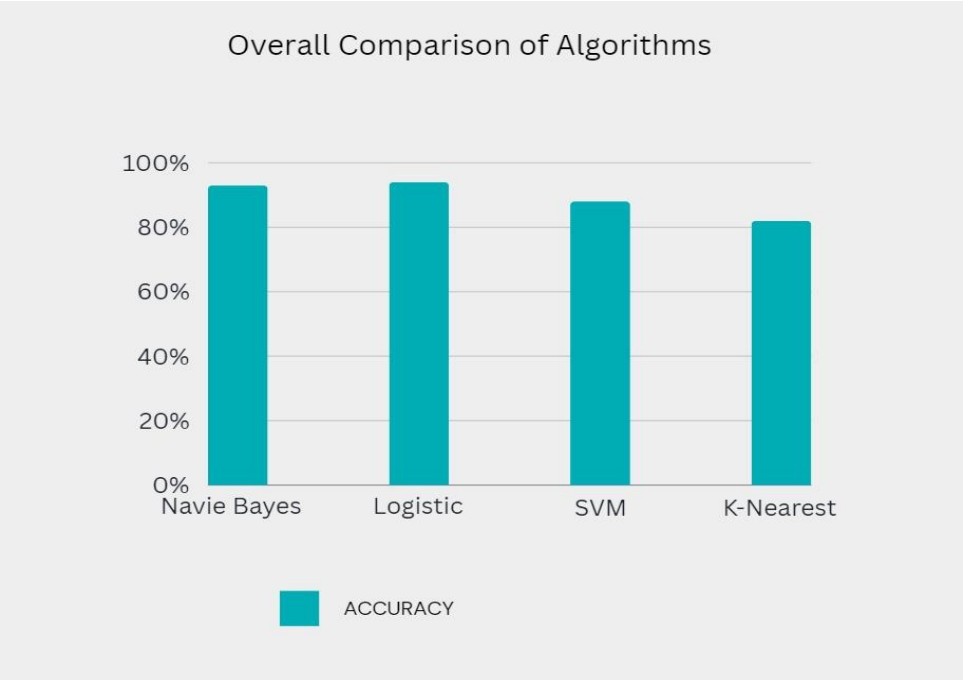

Fig.5.4.5 Output for the Overall Comparison of Algorithms

Algorithm	Accuracy
Navie Bayes Algorithm	0.93
Logistic Algorithm	0.94
SVM Algorithm	0.88
K-Nearest Neighbor Algorithm	0.82

Table.5.4.6 Overall comparison of Algorithms

Table 5.4.6 illustrates the accuracy values for the various algorithms. From the above table we can depict that logistic Algorithm gives the highest accuracy values so we can use this algorithm to predict the sentiment of the model.

Finally, By using Logistic Algorithm we can predict the sentiment of the model by using confusion matrix that ,the model has positive reputation .

6. CONCLUSION AND FUTURE SCOPE

CONCLUSION

Our innovative reputation system evaluates items and their aspects based on online opinions, distinguishing itself with cross-platform compatibility and advanced spam filtering. It utilizes a state-of-the-art aspect-based sentiment analysis model, presenting reputation values through comprehensive visualizations. Outperforming four existing systems in both user and expert evaluations, our roadmap involves expanding capabilities to generate textual attribute summaries and adapting for multilingual content. This ensures enhanced versatility and broad applicability, positioning our system as a leading solution for nuanced and reliable online reputation assessments

FUTURE SCOPE

The future scope of our innovative reputation system involves refining and expanding adaptability by integrating natural language processing advancements and advanced machine learning models. Incorporating user feedback mechanisms will enhance sentiment analysis and reputation assessments. The system's multiplatform compatibility will extend to emerging online spaces, ensuring comprehensive evaluations. Future plans include integration with augmented and virtual reality for immersive user experiences. Collaboration across diverse industries aims to validate and deploy the system, solidifying its role as a universally applicable tool for informed decision-making based on aggregated online opinions.

REFERENCES

[1] Benlahbib; El Habib Nfaoui"Cross-Platform Reputation Assessment Achraf Boumhidi; Abdessamad " IEEE Access ;volumme 10

[2] Susanti Gojali; Masayu Leylia Khodra Aspect-Based sentiment analysis for review rating prediction 2016 International Conference On Advanced Informatics: Concepts, Theory And Application (ICAICTA)

[3] Wenxuan Shia, Maoqiang XiebAddressing Spam in Reputation Evaluation ,published in 2013 by AASRI Conference on Parallel and Distributed Computing Systems

[4] Holly Esquivel ,Aditya Akella ,Tatsuya MoriOn the effectiveness of IP reputation for spam filtering ,IEEE access by Communication Systems and Networks (COMSNETS), 2010 Second International Conference